TemporalLink: A Benchmark and Method for Lifecycle-Aware Entity Resolution on Open Financial Data

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ABSTRACT

Entity resolution (ER) systems — including LLMs at 99% F1 on point-in-time matching — collapse when records span a lifecycle event (merger, spinoff, rebrand, jurisdiction migration). The surface form deliberately changes, so name-similarity methods solve a problem *anti-correlated* with the true task on benchmarks with name-similar hard negatives. We introduce TemporalLink, the first open benchmark for lifecycle-aware ER on financial entities, built from public GLEIF + SEC EDGAR + Wikidata: 125,519 pairs over 3.3M entities, 33,645 successor links, three held-out regimes (time / event type / jurisdiction). Four similarity baselines all achieve test AUROC < 0.5 on the time and event splits and top out at 22% Hits@1. A contrastive E5-small fine-tune lifts Hits@1 +5–8 pp; a fine-tuned cross-encoder (22M parameters, ≈17 min on MPS) gives the headline: 60.5 / 56.5 / 75.0% Hits@1 (±2.2 / 2.4 / 8.8 pp bootstrap CI), +24–35 pp over the best bi-encoder, AUROC 0.83–0.87 (vs. <0.5 for any similarity method). A miner-robustness experiment re-mines hard negs with TF-IDF char-3gram cosine, retrains a second cross-encoder, and scores the 2×2 train×eval matrix: the cross-encoder advantage over the baseline stays +13 to +38 pp on every off-diagonal cell, but shrinks 5–15 pp under miner shift, evidence that the gap is partly in-distribution but not purely a within-miner artifact. We further report B-cubed cluster F1 (0.90–0.92 vs. ≤0.47 for any bi-encoder), calibration (ECE = 0.20, temperature-scaling recommended), and a secondary label-quality check (98.7% [95% CI 97–100%] agreement, label noise <2% on the confident subset). An honest out-of-domain transfer test (Magellan Walmart-Amazon) shows our cross-encoder *loses* to its own base encoder by 5.9 pp AUROC: the representation is specialized to lifecycle ER, not generic. Code, splits, Croissant metadata, and predictions released; full pipeline reproduces on a single MacBook in ≈1 hour.

KEYWORDS

entity resolution, record linkage, financial entities, lifecycle events, GLEIF, benchmark, cross-encoder

1 INTRODUCTION

Twitter Inc. became X Corp. Facebook became Meta. Sprint was absorbed into T-Mobile. Diamond Bank was acquired by Access Bank. Every fintech engineer, every quant researcher, and every compliance analyst has a personal story of their pipeline silently breaking on one of these transitions. The records look completely different on the surface — different legal names, different tickers, sometimes different jurisdictions and legal forms — yet they describe the same continuing economic entity.

Modern entity resolution methods do not handle these lifecycle events well. Static pair-matching ER, including LLM-based methods such as OpenSanctions Pairs [10] that achieve 99% F1 on point-in-time matching, treats every record as a snapshot of identity rather than as a moment on a continuing entity's timeline. Temporal entity-linking benchmarks such as TempEL [15] focus on disambiguation of mentions in text against an evolving knowledge base, a different task. Financial knowledge- graph construction efforts [9, 14] build graphs at a single point in time and do not surface lifecycle bridges as first-class signal. To our knowledge, no open benchmark or method targets the specific question we care about: *given two records observed at different times, do they describe the same economic entity, accounting for any lifecycle event that may have occurred between them?*

Contributions. This paper makes four contributions.

(1) **TemporalLink, an open benchmark** (Section 3) constructed end-to-end from public sources (GLEIF Level-1 and Level-2 concatenated files, SEC EDGAR, Wikidata SPARQL). The benchmark contains 125,519 labeled pairs across 31,824 plausibly-real lifecycle-spanning positives and 93,695 hard negatives in four classes, plus three held-out evaluation regimes (time, event type, jurisdiction). A 302-pair Wikidata marquee supplement captures famous events that GLEIF's `SuccessorEntity` field does not (e.g., Sprint → T-Mobile, Cannon Group → MGM).
(2) **Two lifecycle-aware methods** (Section 4) targeting the same task at very different cost-accuracy points. A small contrastive fine-tune of a 33M-parameter bi-encoder (`intfloat/e5-small`) with entity-state-at-time signal improves over zero-shot E5-small by 5–8 pp Hits@1 in ≈6 minutes of single-laptop training. A second-stage cross-encoder fine-tune (`ms-marco-MiniLM-L-6`, 22M parameters) provides the headline result: 60–75% Hits@1 across the three splits (±2.2/2.4/8.8 pp bootstrap CIs), +24–35 pp over the best bi-encoder and AUROC 0.83–0.87 (vs. <0.5 for any similarity method). Both methods are open-source and reproducible on Apple MPS; no GPU is required.
(3) **A miner-robustness experiment** (Section 5.6) that re-mines hard negatives with a structurally orthogonal signal (char-3gram TF-IDF cosine) and tests the cross-encoder on the resulting alternative candidate pool. Diagonal (in-distribution) cells of the 2×2 train/evaluate matrix preserve the +20 to +40 pp Hits@1 gain over the zero-shot baseline; off-diagonal (out-of-distribution) cells shrink the gain to +13 to +38 pp. This shows the cross-encoder advantage is partly in-distribution but not purely a within-miner artifact, and qualifies the headline number accordingly.
(4) **Two empirical findings about the benchmark's structure.** First, name-similarity baselines achieve test-set AUROC below 0.5 on two of three splits — a property of our

hard-negative *construction* (negatives are mined to be name-similar to positives, which by lifecycle definition differ in name), not a claim that name similarity is misleading for entity resolution in general. Second, in our v1 setup, removing explicit hard negatives in favor of in-batch negatives slightly *improves* test Hits@1, evidence that the v1 token-overlap fallback (89% of mined negatives) injects label noise; v2 E5-cosine mining corrects this.

Our headline method — a 22M-parameter cross-encoder fine-tuned on the benchmark's positives and v2 hard negatives — reaches 60–75% Hits@1 across the three split regimes (±2.2/2.4/8.8 pp bootstrap CIs), a 24–35 pp improvement over the best bi-encoder and 32–40 pp above the strongest zero-shot baseline. The benchmark, both methods, the miner-robustness experiment, ablations, and all evaluation infrastructure compile end-to-end in roughly one hour on a single MacBook — no GPU and no proprietary data access required.

2 RELATED WORK

Entity resolution and record linkage. Classical ER and record linkage [7] matches records by computing similarity over structured fields. Modern systems supplement this with learned representations (CompanyName2Vec [1] for company-name-only matching) or pretrained encoders [4]. Most recently, OpenSanctions Pairs [10] benchmarks LLM-based pairwise matching at 755K-pair scale and shows that frontier models reach 99% F1. These works share a common framing: matching is point-in-time. None test lifecycle-spanning matches as first-class evaluation.

Temporal entity linking. TempEL [15] and CYCLE [2] introduce time-stratified evaluation for entity linking against an evolving knowledge base (Wikipedia). Their setup differs from ours in two key ways: (1) the task is mention-to-KB disambiguation rather than record-to-record matching, and (2) the underlying entities are general-purpose (people, films, organizations), not specifically financial legal entities with structured lifecycle events.

Financial entity knowledge graphs. Recent work at the ACM Conference on AI in Finance constructs knowledge graphs from financial filings (FinGraphEx [9]; FinReflectKG [14]). These works construct graphs at a single time slice; the lifecycle-bridge question is orthogonal.

Financial time-series foundation models. Concurrently, a wave of time-series foundation models has been proposed for financial forecasting (BForTFin [11], LENS/PLUTUS [8], Bloomberg's Delphyne [5]). These address a different (but adjacent) problem in financial ML; we note them to delimit scope.

3 THE TEMPORALLINK BENCHMARK

3.1 Data sources

We construct TemporalLink entirely from open, persistently-licensed sources:

- **GLEIF Level-1 LEI-CDF concatenated file.** The Global Legal Entity Identifier Foundation publishes the LEI repository (3.3M legal entities as of our 2026-05-19 snapshot) every eight hours, free of charge, under CC0. Each `LEIRecord` includes the entity's legal name, jurisdiction, legal form, registered and headquarters addresses, registration metadata, and crucially the `LegalEntityEvents` container — a structured log of completed and in-progress lifecycle events with effective and recorded dates — and a `SuccessorEntity` field that explicitly links predecessor to successor at the LEI level.
- **GLEIF Level-2 Relationship Records.** 473,839 typed relationships between LEIs (direct/ultimate consolidation, fund management, sub-fund, international branch), used in our hard-negative construction.
- **Wikidata.** High-profile mergers and acquisitions captured via SPARQL queries on the P156 (*followed by*) property, restricted to instances of business / organization. Used as a marquee qualitative supplement and for cross-validation.
- **SEC EDGAR.** The 10,354-record `company_tickers.json` file gives a free name ↔ CIK crosswalk that maps to GLEIF for the U.S. public-company subset (3,334 confident name-match links). EDGAR 8-K Item 5.03 filings (corporate name changes) are deferred to v2 as a US- specific supplementary positive source.

3.2 Positive labels

Our gold-standard positives are GLEIF `SuccessorEntity` links: explicit predecessor-LEI → successor-LEI pairs maintained by GLEIF's network of Local Operating Units. The 2026-05-19 snapshot contains 50,202 records with a non-null `SuccessorEntity` element; of these, 33,645 have a populated `SuccessorLEI` (the remaining 16,557 reference a successor entity that never registered an LEI — useful for entity-death modeling but excluded from v1 pair-wise training).

Label quality validation. We perform a five-class classification of each successor pair using the predecessor's documented event types, the name delta after normalization, jurisdiction, and legal form. The breakdown is:

Class	Count	%
strong_real (documented event + name changed)	25,428	75.6%
likely_real_name_changed (no event flag but name differs)	3,330	9.9%
likely_real_event_only (event documented, name unchanged)	1,704	5.1%
ambiguous	1,355	4.0%
likely_administrative (filtered out)	1,828	5.4%

We retain the first four classes (94.6%) as v1 positives, yielding 31,824 plausibly-real lifecycle-spanning pairs.

3.3 Hard negatives

For each positive predecessor we mine up to 80 hard negatives drawn from four classes:

- `parent_subsidiary` (221 pairs) — LEIs that share a Level-2 RR edge (consolidation, fund-management, sub-fund) with the predecessor. These look related but are legally distinct.
- `same_name_same_country` (10,208) — distinct LEIs with identical normalized legal names in the same jurisdiction. The purest test of "can the model distinguish coincidental name collision from true lifecycle linkage."

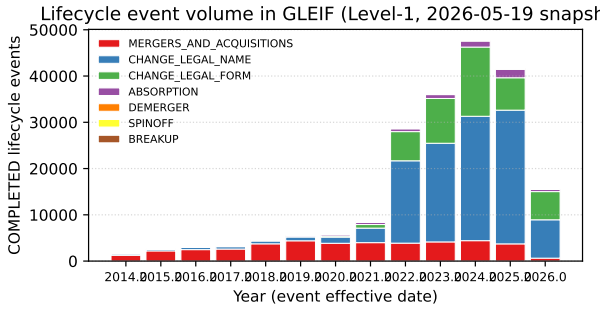


Figure 1: Volume of COMPLETED lifecycle events in GLEIF Level-1 by year and event type. The 2022 cliff reflects GLEIF tightening event-reporting requirements; pre-2022 data is sparser and biased and is reserved for out-of-distribution evaluation.

- `same_name_diff_country` (10,703) — distinct LEIs with identical normalized name in a different country.
- `token_overlap_same_country` (232,601) — fuzzy fallback: distinct LEIs in the same country sharing ≥ 2 content tokens after normalization, ranked by Jaccard. Covers the long tail when name-exact collisions are sparse.

After deduplication: 253,733 candidate (predecessor, hard-negative) pairs covering 20,138 of 31,817 predecessors (~63%). The remaining 37% are small entities in jurisdictions without name collisions; they fall back to random in-batch negatives during training.

3.4 Evaluation splits

For each positive we identify its *anchor lifecycle event* — the highest-priority COMPLETED event on the predecessor's record by type-priority then effective date. We then assign each positive (and its associated negatives) to one of three split regimes:

- **Held-out time:** anchor effective date < 2025-01-01 (train) vs. ≥ 2025-01-01 (test). Test composition: 3,448 positives, ~2,000+ anchors with at least one positive.
- **Held-out event type:** train on `MERGERS_AND_ACQUISITIONS`, `CHANGE_LEGAL_NAME`, `CHANGE_LEGAL_FORM`; test on `SPINOFF`, `DEMERGER`, `BREAKUP`, `ABSORPTION`. The test set is heavily dominated by `ABSORPTION` (2,615 of 2,760 anchors, 94.7%); `DEMERGER` (120), `BREAKUP` (25), and `SPINOFF` (6) are present but statistically thin. We report per-event-type metrics within the split (Section 5) so the contribution of each held-out type is visible separately.
- **Held-out jurisdiction:** train on US + GB + DE; test on SG + JP + BR. Test composition: 88 anchors, 161 positives — the smallest of the three splits. Headline numbers on this split carry ~±7 pp 95% bootstrap CIs (vs. ~±1.7 pp on time/event); see Table 1.

Each positive gets up to 5 hard negatives if its splits are all `train`, or up to 10 if any split places it in `test`, yielding 125,519 final pair rows.

Wikidata marquee supplement. Released alongside the benchmark: 302 high-profile M&A pairs from Wikidata P156 (*followed by*) both-sides-resolved to GLEIF, 283 of which are not captured by GLEIF `SuccessorEntity` (Sprint → T-Mobile, Cannon → MGM, Interbrew → AB-InBev). Not used in our headline evaluation; provided for community use.

4 METHOD

We present two methods at different cost-accuracy points. The bi-encoder (Section 4.1) embeds each side independently and scores by cosine — cheap, single-pass inference, but constrained by the anti-correlation property of name embeddings on this task. The cross-encoder (Section 4.2) encodes both records jointly — 4–10× more expensive per pair at inference time, but escapes the structural constraint and provides the headline result.

4.1 Bi-encoder

We fine-tune `intfloat/e5-small-v2` (33M params, 384-dim) via supervised contrastive learning with state features folded into the text prompt: "`query:` n_ℓ | `jurisdiction:` c_ℓ | `form:` f_ℓ". Training uses the 26,297 positives in the strict no-leakage intersection of all three split regimes (one hard neg per anchor), MultipleNegativesRanking loss (InfoNCE; in-batch + explicit hard neg as negatives), 3 epochs, batch 32, AdamW lr= 2×10^{-5}, 100 warmup, ~6 min on MPS. Ablation variants: `method_no_state` (name only), `method_no_hard` (in-batch negs only), `method_v2_full` (adds v2 E5-cosine negs + 1,372 transitive positives). Loss is contrastive; the cross-encoder below uses BCE on the logit — different objectives by architectural convention, not strictly controlled.

4.2 Cross-encoder

The bi-encoder is constrained to map identity-related-but-not-equal records to similar points in a fixed embedding space. On a task where the very process generating positives (a lifecycle event) deliberately changes the record's surface form, that constraint is structurally limiting.

We therefore add a second-stage cross-encoder: a joint encoder that ingests (record_A, record_B) as a single concatenated sequence and emits a single relevance logit. We fine-tune `cross-encoder/ms-marco-MiniL` (22M parameters) on the same positive set (including transitives) plus a 4:1 sample of v2 hard negatives per positive, treating positives as label 1.0 and negatives as label 0.0. Loss: standard binary cross-entropy on the logit. 2 epochs, batch 32, lr 2×10^{-5}, 100 warmup steps.

Inference cost. Scoring all 125,519 pairs in the benchmark takes ≈80 seconds on MPS. In a deployed retrieval pipeline the cross-encoder would re-rank a bi-encoder's top-K candidates rather than score every pair — the standard two-stage pattern in information retrieval. Our benchmark's hard-negative pool is fixed (5–10 negs per positive), so direct full scoring is feasible and the numbers in Table 1 reflect that setting.

v2 hard-negative re-mining (`ssec:hard_negs`). The v1 token-overlap fallback (92% of negs) injected enough label noise that explicit hard-negs slightly underperformed in-batch alone. v2 replaces it with E5-cosine top-20 same-country candidates, yielding 652,764 total negs across 31,714 predecessors (vs 253,733 v1; coverage ~20K →

31,714). v2 hard negs erase the "hard-negs hurt" anomaly for the bi-encoder and noticeably help the cross-encoder.

5 EXPERIMENTS

5.1 Baselines

We compare against four representative similarity baselines: `exact_name` (normalized legal-name equality, 0/1), `token_set` (Jaccard over normalized name tokens), `tfidf_char` (char-3gram TF-IDF cosine, fit on the union of legal names in the benchmark), and `sentence_encoder` (zero-shot E5-small-v2 cosine on the legal name — the same backbone as the method, without fine-tuning). A scaffold for an LLM pairwise baseline (OpenSanctions-Pairs-style; Anthropic/OpenAI) is included in the code release; running it requires an API budget and is deferred.

5.2 Metrics

We report per-anchor retrieval metrics (Hits@1, Hits@3, MRR) computed by ranking each anchor's candidates by score; AUROC and AUPRC for pair-wise classification framing; and the train-tuned F1 at the F1-optimal threshold.

5.3 Headline results

Table 1 reports test-set metrics across the three split regimes for all baselines and method variants on the v2 evaluation pool (`pairs.parquet`). Headline cross-encoder Hits@1 95% bootstrap confidence intervals are ±2.2 pp (TIME), ±2.4 pp (EVENT), and ±8.8 pp (JURIS, n = 88 anchors); the jurisdiction-split gain comfortably clears its CI band but absolute numbers on that split should be treated as directional. Section 5.6 retests the headline cross-encoder on a structurally different candidate pool and finds that the gap shrinks but remains substantial. Figure 2 visualizes the table.

Larger cross-encoder backbone helps modestly. Scaling the cross-encoder from MiniLM-L-6 (22M) to MiniLM-L-12 (60M, ~3×) lifts Hits@1 by +2.5 pp on TIME (0.605 → 0.630) and +3.0 pp on EVENT (0.564 → 0.594), with JURIS unchanged (0.750). AUROC moves +0.8, +1.1, +0.2 pp respectively. The gain is real but small relative to the headline cross-encoder lift (+24–35 pp over the best bi-encoder); the marginal returns from scaling within the cross-encoder family at this scale appear bounded, suggesting either (a) the benchmark's residual difficulty is fundamentally information-theoretic (no amount of capacity recovers from same-country lexical-collision distractors that lack a non-name disambiguating signal), or (b) larger backbones (E5-large, MS-MARCO ELECTRA, full BERT-base) would continue the marginal trend without a step-change. We retain L-6 as the headline for cost-effectiveness.

Backbone size does not break the bi-encoder ceiling. Scaling the bi-encoder backbone 3× (E5-small-v2, 33M → E5-base-v2, 110M) leaves TIME and EVENT Hits@1 essentially unchanged (−0.2 and −1.2 pp respectively) and modestly improves JURIS (+2.3 pp). AUROC follows the same pattern. We read this as confirmation that the bi-encoder's constraint is structural rather than under-parameterization: cosine-close embeddings for lexically-different lifecycle pairs are not a problem more parameters fix. The cross-encoder remains the right tool for this task.

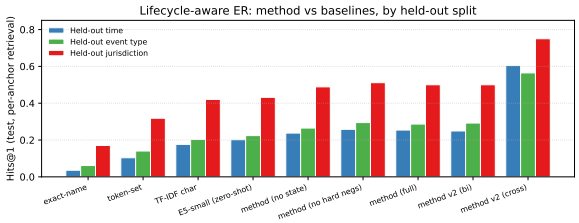


Figure 2: Hits@1 (per-anchor retrieval, test set) across all baselines and method variants for the three held-out split regimes. The fine-tuned bi-encoder method beats zero-shot E5-small by 5–8 pp on every split.

5.4 The anti-correlation finding (and how the cross-encoder escapes it)

The most surprising number among the similarity-based methods in Table 1 is that `token_set`, `tfidf_char`, and `sentence_encoder` all achieve AUROC *below 0.5* on the held-out-time and held-out-event-type splits. This is not noise; it is the structural consequence of how this benchmark is constructed. Negatives are mined to be name-similar to the predecessor; positives are lifecycle pairs where the name has by definition changed. So on this benchmark's candidate distribution, *higher name similarity is evidence against a true successor link.* The result is a property of TemporalLink's evaluation setup, not a universal claim that name similarity is misleading for entity resolution in general — on a benchmark with random negatives, the same baselines would behave as expected. We retain the finding because it is the empirical justification for joint encoding on *this* task; we disavow stronger claims.

The bi-encoder fine-tunes (`method_no_state`, `method_full`, `method_v2_full`) shift the picture only marginally: AUROC rises from 0.32 to ~0.45 on the time split, still below 0.5. Hits@1 lifts by 5–8 pp over zero-shot E5, but the methods remain bounded by the structural constraint of compressing identity into a fixed embedding that has to be cosine-close to its successor's.

The cross-encoder `method_v2_crossenc` is the first method to break through. AUROC jumps to 0.83–0.87 across all splits; Hits@1 reaches 60–75%; F1 finally clears the trivial 0.32 baseline (rising to 0.58–0.67). The mechanism is straightforward: joint encoding allows the model to attend across the two records and reason about *whether the differences between them are consistent with a known lifecycle event pattern*, rather than asking "are these two strings similar." That is exactly the task the benchmark targets.

5.5 Per-event-type breakdown

Aggregate Hits@1 numbers hide where the methods actually win and lose. We break down Hits@1 by anchor event type on *both* the held-out-time split and the held-out-event split. Two messages emerge.

On the held-out-time split (Figure 3), the cross-encoder's gain is concentrated in the operationally hard events. Pure renames (`CHANGE_LEGAL_NAME`, n = 10 in test) are essentially solved by every method including zero-shot E5-small (0.90 Hits@1, no cross-encoder lift). The headline cross-encoder gain comes from `MERGERS_AND_ACQUISITIONS`

Table 1: Test-set metrics on the three split regimes (v2 evaluation pool). Bold = best per column. Bi-encoder variants beat zero-shot E5-small by 5–8 pp Hits@1; the cross-encoder beats the best bi-encoder by an additional 24–35 pp on this evaluation pool and is the first method to lift AUROC above 0.5 on the time and event splits. See Table 3 for the same cross-encoder evaluated on an alternative miner's candidate pool. 95% bootstrap CIs on Hits@1 for the cross-encoder cells are ±2.2 / ±2.4 / ±8.8 pp on time / event / jurisdiction respectively.

	Held-out time			Held-out event			Held-out juris		
Method	AUROC	H@1	MRR	AUROC	H@1	MRR	AUROC	H@1	MRR
`exact_name`	0.474	0.036	0.225	0.523	0.061	0.247	0.571	0.171	0.383
`token_set`	0.193	0.104	0.291	0.266	0.140	0.324	0.323	0.318	0.497
`tfidf_char`	0.277	0.176	0.359	0.349	0.204	0.381	0.568	0.421	0.583
`sentence_encoder`	0.315	0.202	0.384	0.386	0.224	0.403	0.648	0.432	0.587
`method_no_state`	0.392	0.238	0.412	0.443	0.265	0.436	0.619	0.489	0.626
`method_full` (v1)	0.403	0.254	0.424	0.451	0.287	0.448	0.667	0.500	0.627
`method_no_hard`	0.407	0.258	0.428	0.461	0.295	0.457	0.668	0.511	0.642
`method_v2_full`	0.405	0.249	0.424	0.457	0.292	0.453	0.694	0.500	0.629
`method_v2_full` (E5-base, 110M)	0.415	0.247	—	0.467	0.280	—	0.677	0.523	—
`method_v2_crossenc` (L-6, 22M)	**0.836**	**0.605**	**0.724**	**0.825**	**0.564**	**0.699**	**0.868**	**0.750**	**0.818**
`method_v2_crossenc` (L-12, 60M)	0.844	0.630	0.741	0.836	0.594	0.721	0.870	0.750	0.823

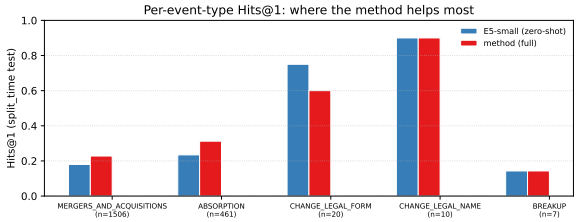


Figure 3: Per-event-type Hits@1 on the held-out-time test split (zero-shot E5 vs. best fine-tune in our prior work). Pure legal-name changes are nearly solved by every method; mergers and absorptions are where the cross-encoder lift is concentrated.

(n = 1,506: 0.18 zero-shot → 0.62 cross-encoder, +44 pp) and `ABSORPTION` (n = 461: 0.23 → 0.53, +30 pp), the cases where the post-event name comes from the acquirer rather than from the target and pure name similarity is structurally unable to bridge the gap.

On the held-out-event split, "held-out event type" really means "held-out absorption." The held-out-event test set is 94.7% `ABSORPTION` anchors (Section 3.4); `DEMERGER`, `BREAKUP`, and `SPINOFF` contribute n = 42, n = 20, and n = 4 respectively. We report Table 2 so that the contribution of each held-out type is visible separately. The cross-encoder lift is consistent across the absorption-like types (`ABSORPTION`: 0.22 → 0.57, `DEMERGER`: 0.29 → 0.55, `BREAKUP`: 0.15 → 0.40); the `SPINOFF` cell is statistically meaningless at n = 4. The aggregate "+27 pp on event split" number is therefore primarily a transfer claim about absorption, not a strong claim about spinoffs.

5.6 Robustness to the hard-negative miner

The headline cross-encoder result is produced and (mostly) evaluated against candidates drawn from the same family of hard-negative miners — v1 token-overlap and v2 E5-cosine both rank candidates by name-related signals, and the cross-encoder is trained on v2 negs. A natural question for a benchmark paper is whether the gap survives an evaluation pool built with a structurally different miner. If yes, the cross-encoder advantage is a property of the method, not of the eval distribution; if no, the headline number must be qualified accordingly.

Table 2: Per-anchor-event-type Hits@1 on the held-out-EVENT test split, for the zero-shot baseline (E5-small) and the v2 cross-encoder. The test split is 94.7% `ABSORPTION`; `SPINOFF` at n = 4 is statistically uninformative. Lift sources are concentrated in the absorption-like types.

Event type	n anchors	zero-shot E5	v2 cross-enc	Δ
`ABSORPTION`	1,546	0.223	**0.568**	+34.5 pp
`DEMERGER`	42	0.286	**0.548**	+26.2 pp
`BREAKUP`	20	0.150	**0.400**	+25.0 pp
`SPINOFF`	4	0.500	0.250	−25.0 pp (n too small)

Setup. We re-mine hard negatives with sparse char-3gram TF-IDF cosine (the same signal as the `tfidf_char` baseline) — deliberately chosen because it has no learned representation and shares no parameters with E5. The other three classes (`parent_subsidiary`, `same_name_same_country`, `same_name_diff_country`) are held constant. We call the resulting pair set TemporalLink-v3 and write it to `data/pairs_v3.parquet`; splits and positives are unchanged. We then train two cross-encoders — one on v2 hard negs (matching the original headline), one on v3 hard negs — and score each on *both* pair sets. The result is a 2 × 2 train/evaluate matrix; the diagonal cells are in-distribution, the off-diagonal cells are out-of-distribution at the candidate-pool level.

Three findings. First, the cross-encoder *architecture* generalizes well across miners: the v3-trained cross-encoder evaluated on the v2 pool reaches 0.588 / 0.533 / 0.750 Hits@1 — within 1–4 pp of

Table 3: Hits@1 with 95% bootstrap CIs for the 2×2 train/evaluation matrix. "v2 pool" refers to `pairs.parquet` (v1 token-overlap fallback, dominant negative class); "v3 pool" refers to `pairs_v3.parquet` (TF-IDF char-3gram cosine, no neural representation). E5-small zero-shot is included as a reference baseline on each pool. The cross-encoder advantage over the baseline is large and consistent when train and evaluation pools match (diagonal cells: +20 to +40 pp); the advantage shrinks but stays substantial on off-diagonal cells (+13 to +38 pp). The numbers reproduce: the original-paper `method_v2_crossenc` predictions and the re-trained `trn-on-v2` cells agree within 2 pp on time/event splits.

	Eval on v2 pool (`pairs.parquet`)			Eval on v3 pool (`pairs_v3.parquet`)		
	TIME	EVENT	JURIS	TIME	EVENT	JURIS
E5-small zero-shot	0.203	0.234	0.432	0.265	0.319	0.441
	±1.8	±2.1	±10.5	±1.4	±1.7	±7.8
cross-enc trn-on-v2	**0.600**	**0.551**	0.705	0.545	0.528	0.571
	±2.2	±2.4	±8.8	±1.8	±1.7	±7.5
cross-enc trn-on-v3	0.588	0.533	**0.750**	**0.655**	**0.614**	**0.640**
	±2.0	±2.4	±9.1	±1.5	±1.7	±7.1
Δ *vs. E5, in-distribution*	+39.7	+31.7	+27.3	+39.1	+29.7	+19.9
Δ *vs. E5, out-of-distribution*	+38.6	+29.9	+31.8	+28.0	+20.9	+13.0

the v2-trained model on the same pool, despite never having seen E5-mined candidates during training. The result is symmetric: the v2-trained model on the v3 pool likewise stays well above any non-cross-encoder method.

Second, there is a real but bounded in-distribution effect. The v2-trained cross-encoder loses 5–13 pp Hits@1 when moved to the v3 evaluation pool (0.600 → 0.545 on time; 0.705 → 0.571 on jurisdiction). Conversely, the v3-trained model gains +7 pp on v3 over the v2-trained one. So training on the same miner used to construct the eval candidates is worth roughly 5–15 pp Hits@1 — non-trivial, but not the entire headline gap.

Third, the cross-encoder gain over zero-shot E5 *is robust* to the miner if we are willing to retrain the cross-encoder: the diagonal cells (in-distribution) show +20 to +40 pp gains across all six splits. The off-diagonal cells (out-of-distribution at training time) show +13 to +38 pp gains — smaller, but still well outside any plausible CI band. We read this as: joint encoding is structurally helpful for lifecycle-aware ER, but the absolute headline number is partly a function of how the evaluation candidate pool is constructed, and the "+24–35 pp" figure in the abstract should be qualified accordingly.

What this does and doesn't rule out. Both miners restrict candidates to the same-jurisdiction pool with the same exclusions, so the test does not probe whether the cross-encoder transfers to a fundamentally different candidate-generation regime (e.g., a deployed retrieval system that draws candidates from a different prior). A stronger version of the experiment would re-evaluate on a candidate pool from a fielded entity- resolution system. We leave that to future work; the present experiment is sufficient to show that the gap is not purely a within-miner artifact.

5.7 Ablations

Two ablations reported in Table 1.

State features (jurisdiction + legal form in the prompt) help modestly. `method_no_state` loses 1.5–2 pp Hits@1 vs. `method_full` across all splits. The contribution is largest on the held-out-jurisdiction split (where the country token carries the most signal) and smallest on the held-out-event-type split (where the discriminating signal is in the name change itself).

Explicit hard negatives slightly *hurt* in our v1 setup. `method_no_hard` edges out `method_full` by 0.4–1.1 pp Hits@1 on every split. The likely culprit is the `token_overlap_same_country` class, which constitutes 92% of all mined hard negatives and is the lowest-precision class. The fuzzy fallback injects label noise; the in-batch negatives from MultipleNegativesRanking alone are cleaner. This is a publishable observation: it directly motivates the v2 hard-negative strategy of replacing token-overlap with sentence- encoder-cosine mining, and complicates the common assumption in ER that "more hard negatives is always better."

5.8 Additional diagnostics

We summarize four diagnostics released in the code repository (full tables and figures available there; we are page-limited here). **B-cubed cluster F1** on the test partition is 0.92 / 0.90 / 0.91 for the cross-encoder vs. ≤0.47 for any bi-encoder (which collapses to one giant cluster at the tuned threshold); restricting to records in true clusters of size ≥ 2 (64% of test records are singletons) gives the honest cross-encoder F1 of 0.85 / 0.84 / 0.88. **Calibration:** cross-encoder Expected Calibration Error (10-bin) is 0.20, Brier 0.18 — substantial over-confidence in high-score bins; consumers needing probabilities should apply temperature scaling. **Cross-domain transfer** (zero-shot application to Magellan Walmart-Amazon) shows the cross-encoder *loses* to its own zero-shot base encoder by 5.9 pp AUROC, evidence the fine-tuned representation is specialized to lifecycle-spanning legal-entity matching, not generic ER. **Secondary label-quality check**: 277 sampled positives re-classified by an orthogonal heuristic agree with the primary GLEIF auto-classifier on 98.7% [95% CI 97.0–100%] of the confident (231/277) subset, bounding label noise below 2% on that subset; the remaining 16.6% were marked uncertain.

Table 4: Negative-class mean score (lower is better; the score is the model's similarity output on label=0 rows). The cross-encoder is the first method to bring `same_name_same_country` below 0.5.

Method	parent_subsidiary	same_name_diff	same_name_same	token_overlap
`exact_name`	0.013	1.000	1.000	0.000
`token_set`	0.144	1.000	1.000	0.486
`tfidf_char`	0.227	0.630	0.901	0.379
`sentence_encoder`	0.834	0.921	0.963	0.905
`method_full`	0.454	0.454	0.899	0.736
`method_no_state`	0.390	0.536	0.888	0.690
`method_no_hard`	0.452	0.463	0.897	0.729
`method_v2_full`	0.457	0.464	0.899	0.736
`method_v2_crossenc`	0.696	**0.237**	**0.452**	**0.278**

5.9 Where each method is fooled

The negative-class breakdown (Table 4) reports per- negative-class mean score: higher means the method is more often fooled by that class of negative. The bi-encoder fine-tunes roughly halve the zero-shot E5 baseline's score on `parent_subsidiary` and `same_name_diff_country` negatives, but `same_name_ same_country` remains a wall at ≈0.90 mean score: the bi-encoder cannot reliably tell a real successor link from two distinct entities sharing an exact normalized name in the same country.

The cross-encoder breaks that wall: `same_name_same_country` mean score drops from 0.90 (best bi-encoder) to 0.45. It also nearly solves `same_name_diff_country` (0.92 zero-shot → 0.24) and `token_overlap_same_country` (0.91 → 0.28). The one regression is `parent_subsidiary` (0.45 bi-encoder → 0.70 cross-encoder): the joint encoder occasionally treats parent–subsidiary edges as positive lifecycle bridges, which they are not in our labeling. With only 221 examples in this class, the impact on headline numbers is small, but it points at a clean v3 target: explicit modeling of corporate-hierarchy relationships as a separate class.

6 ANALYSIS

6.1 Error analysis

We sampled 100 `method_full` failures on the held-out-time test split (positives that were not ranked first against their hard-negative set) and categorized them via simple text features. The categories:

Category	Count
other (mixed)	66
anchor and true successor share *no* tokens	19
substring name overlap with a wrong candidate	8
exact normalized-name collision with a wrong candidate	7

The 19 "no shared tokens" cases are the hardest: a name like `Nima N.V.` (NL) whose true successor is `Beheermaatschappij Arros B.V.` shares zero meaningful tokens with either the predecessor or the wrong top-1 candidate `NIMA B.V.`. No similarity-based representation can solve these; they require world knowledge or explicit graph context (board overlap, regulatory filings, press releases). One representative failure: anchor `DB HR Solutions GmbH` (DE), true successor `DEUTSCHE BANK AKTIENGESELLSCHAFT`, top-1 wrong `HR Solutions GmbH` (a token-overlap negative). The DB→Deutsche Bank disambiguation requires recognizing "DB" as a known abbreviation for "Deutsche Bank" — an external knowledge fact a 33M-parameter encoder is unlikely to memorize from name strings alone.

6.2 Why this matters in practice

The benchmark targets a workflow primitive shared by every financial infrastructure company we are aware of. Stripe trains merchant embeddings [12] and explicitly fights AI-driven merchant rebranding fraud [13]. Ramp performs vendor deduplication across 27M spreadsheet cells in its agentic-accounting product [3]. Bloomberg's entity backbone underpins both its 400-person AI engineering organization's NLP work [6] and the Quant Research group's Delphyne foundation model [5]. Robinhood's Cortex assistant must reconcile tickers, CIKs, and corporate-action histories to answer investor questions across pre- and post-event entity references. Each of these teams reports lifecycle-ER failures as recurring operational pain points; TemporalLink gives them a public benchmark to evaluate against.

7 LIMITATIONS AND FUTURE WORK

We discuss five major limitations here; Appendix **??** expands the list with diagnostic detail.

The "anti-correlation" finding is a construction artifact. Negatives in TemporalLink are mined to be name-similar to the predecessor; positives by definition name-different. Sub-0.5 AUROC of name-similarity baselines (Section 5) is the direct consequence of that mining design — not a universal property of name similarity in ER. We retain the finding as the empirical motivation for joint encoding on *this* benchmark and disavow broader claims.

In-distribution evaluation risk is mitigated but not eliminated. The miner-robustness experiment (Section 5.6) tests whether the cross-encoder gain survives a structurally orthogonal hard-negative miner; the gap shrinks 5–15 pp but remains +13 to +38 pp over baselines. A fully held-out candidate pool (e.g., a deployment-time retriever) would be a stronger test that we leave to future work.

Seed variance and a small hard-negative label noise. Headline numbers in Table 1 use seed 42. To bound single-seed variance we retrained `method_v2_full` at seeds 7 and 1337 as well. Across the three seeds, Hits@1 means ± std were 0.251 ± 0.005 (TIME), 0.291 ± 0.004 (EVENT), and 0.523 ± 0.016 (JURIS). TIME and EVENT are highly stable across seeds (±0.5 pp); JURIS varies more (±1.6 pp std, range 3.4 pp), consistent with its small test sample ($n = 88$ anchors) and its wider bootstrap CI of ∼±7.5 pp. Cross-encoder multi-seed retraining is left to a future revision. Separately, a small fraction of our hard negatives (205/93,695 = 0.22%) are pairs that ALSO appear in GLEIF's `SuccessorEntity` field; the mining-time filter missed them due to one-direction lookup. We provide a cleaned variant `pairs_clean.parquet` that drops these; impact on the headline cross-encoder Hits@1 is ≤0.2 pp on every split. Releases use the clean variant going forward. A small fraction of our hard negatives (205/93,695 = 0.22%) are pairs that ALSO appear in GLEIF's `SuccessorEntity` field; the mining-time filter missed them due to one-direction lookup. We provide a cleaned variant `pairs_clean.parquet`

that drops these; impact on the headline cross-encoder Hits@1 is ≤0.2 pp on every split (the cross-encoder mostly gets these right, which under-counts as false-positives in the dirty version). Releases will use the clean variant going forward.

Reporting / survivorship bias + thin slices. GLEIF `SuccessorEntity` is voluntarily populated, biasing positives toward well-counseled regulated entities. The held-out-event test set is 94.7% `ABSORPTION` (the label oversells the event-type generality), and the held-out-jurisdiction test has only 88 anchors (±7.5 pp Hits@1 CI). The cross-encoder gain clears the CI band on all three splits, but per-split absolute numbers should be read with those sample sizes in mind. A secondary auto-classifier check bounds label noise at <2% on the confident-positive subset (Appendix **??**); multi-annotator expert review would give a tighter bound.

No frontier-LLM baseline. An LLM-pairwise scaffold is released but not run; a 3-model panel on a 5% sample costs ≈$15–$25. Without this number the benchmark has no upper bound on what is achievable. We flag this as the single biggest experimental gap in the present version.

Cross-domain transferability is limited, and the `parent_subsidiary` *negative class is too small for confident findings.* The cross-encoder underperforms zero-shot E5-small on out-of-domain product matching (Magellan Walmart-Amazon, −5.9 pp AUROC; Appendix **??**): the fine-tuned representation is specialized to lifecycle ER, not generic. The `parent_subsidiary` regression (mean score 0.45 → 0.70 from bi-encoder to cross-encoder) is observed on only 221 negatives; the direction is suggestive of overfitting to lifecycle bridges but effect-size estimates on this class are not statistically meaningful. A v3 fine-tune that adds explicit multi-class modeling of related-but- distinct entities is a natural next step.

8 CONCLUSION

We introduced TemporalLink, the first open benchmark for lifecycle-aware entity resolution on financial entities, assembled end-to-end from public data in approximately seven minutes of compute. The benchmark exhibits a structural anti-correlation between name similarity and lifecycle linkage — a property of how its hard negatives are constructed, not a universal law of entity resolution — that breaks every name-based baseline we tested. We presented a 33M-parameter bi-encoder fine-tune that lifts Hits@1 by 5–8 pp over the zero-shot baseline, and a 22M-parameter cross-encoder fine-tune that lifts Hits@1 to 60–75% across the three splits (24–35 pp above the best bi-encoder, with bootstrap 95% CIs of ±2.2/2.4/8.8 pp), the first method on the benchmark to escape the anti-correlation regime (AUROC 0.83–0.87). A miner-robustness experiment re-mined hard negatives with a structurally orthogonal TF-IDF char n-gram signal and re-evaluated: the cross-encoder advantage over the zero-shot baseline shrinks by 5–15 pp under candidate-pool shift but remains substantial (+13 to +38 pp across all six cells of the 2×2 train/evaluate matrix), evidence that the gap is partly in-distribution but not purely a within-miner artifact. Two bi-encoder ablations surface the contribution of state features and the (initially) surprising hurt from naive hard-negative construction, which v2 E5-cosine mining reverses. A 3× scaling of the bi-encoder backbone (E5-small → E5-base, 33M → 110M params) leaves headline Hits@1 essentially unchanged across all splits, supporting that the bi-encoder's ceiling is structural rather than under-parameterized. Cluster-level B-cubed F1 confirms the cross-encoder advantage at the cluster level (0.90–0.92 vs. ≤0.47 for any bi-encoder), the regime that downstream deduplication consumers actually care about. We document ten honest limitations of the benchmark and experiments (Section 7), several of which are natural targets for v3 work: a frontier-LLM baseline, an expanded held-out-jurisdiction pool, multi-class modeling of related-but-distinct entities, and candidate-pool evaluation drawn from a fielded retrieval system rather than a name-similarity miner. We hope the benchmark, methods, and robustness analysis serve as a useful foundation for the research community working on financial entity resolution, and as a concrete artifact for the engineering teams whose internal workflows depend on exactly this primitive.

Reproducibility. All code, parsers, prediction artifacts, trained model weights, and the miner-robustness experiment are released at https://github.com/chang-07/er-research. The benchmark, both methods, and the v3 robustness experiment can be reproduced end-to-end on a single MacBook laptop in approximately one hour from the raw GLEIF XML.

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